

John Nesbit

706-224-5083 | johnmnes06@gmail.com | linkedin.com/in/JohnNesbitCS/ | github.com/JohnNesbit

EDUCATION

Georgia Institute of Technology

Graduating May 2028

Bachelor of Science in Computer Science

Atlanta, GA

Threads: Systems & Architecture and Information Internetworks

GPA: 4.0

Relevant Coursework: Operating Systems, Networking II, Algorithms Analysis, Processor Design, Linear Algebra

EXPERIENCE

Susquehanna International Group

Jun 2026 - Aug 2026

Trading Systems Engineering Intern

Bala Cynwyd, PA

- Architected and built a market-making engine with two other interns in C++ which achieved second in both PnL and overall wire-to-wire latency (p50, p99) across intern teams
- Built a Bayesian intraday volume forecasting model which cut MSE 44% vs. a 30-day EMA intraday volume profile on less liquid names (WING, FIZZ) and 21% on liquid names (LRCX, AAPL)
- Placed second on slippage variance in an intern competition with a C++ VWAP execution algorithm that scheduled fills using a market impact model driven by volume forecasts and short-term price signals

Manhattan Associates

May 2025 - Aug 2025

Software Engineering Intern - Cloud

Atlanta, GA

- Saved \$53k/yr in cloud spend by developing a CI/CD pipeline in C# to resize VMs across multi-tenant groups
- Reduced storage account spend by 16% by automating file purging with a Python Azure Function
- Enabled a F500 retailer to deploy custom business logic by architecting VPN and Azure Function integrations

UGA Center for the Ecology of Infectious Diseases

Jun 2024 - Aug 2024

Research Assistant (REU)

Athens, GA

- Built visualizations of state-level pandemic data in R to support a group modeling vaccination timing; presented findings to 15 researchers

UGA School of Computing

Aug 2023 - May 2024

Research Assistant

Athens, GA

- Co-authored paper on transformer-based time series forecasting in *BigData 2024*
- Unified R and Python (PyTorch, TensorFlow) pipelines into a framework used by 4 graduate projects

PROJECTS

Throughput-Optimized Matching Engine

- Built a frequent-batch-auction exchange in C++23, reaching 1.02M orders/s using batched-syscalls
- Used perf to trace bottlenecks like an HPET clocksource inflating syscall cost and unbalanced softirq handling
- Cut the exchange matching logic to ~2% of CPU time with a batched lock-free SPSC ring buffer
- Benchmarked io_uring against batched-syscalls; benchmarks showed no net throughput gain on this workload

Kaggle ASHRAE III Competition

- Won a silver medal (top 3%) for predicting building energy consumption
- Improved a gradient-boosted tree ensemble via data-cleaning pipelines motivated by permutation-based importance tests in Python

ACTIVITIES

Georgia Tech Formula SAE Electric Racing

Jan 2025 - Apr 2025

DriveBrain Software Engineer

Atlanta, GA

- Added torque controller system to software stack (Server, CAN, Protobuf, Microcontroller) for torque vectoring
- Accelerated development cycles by writing car environment simulation software in C++

TECHNICAL SKILLS

Languages: C++, Python, C

Systems & Tooling: Linux, CMake, Nix, Git, perf, GDB

Libraries: PyTorch, NumPy, Pandas